

Category	RefDes / Item	Qty on Board	Order Qty	Include in Purchase Re	Description	Manufacturer	MPN	Vendor	Vendor Link	Unit Price	Total Price	Notes
No Footprint	JST / crimp contacts	2	10	Yes	JST VH crimp contact for JST Sales America Inc.	SVH-21T-1.1	DigiKey	https://www.digikey.com/en/products/detail/jst-sales-america-inc/SVH-21T-1.1/527386	0.064	0.64	Cable-side part. Order extra contracts for crimping mistakes.	
No Footprint	JST 2-pos housing	1	10	Yes	JST VH 2-position housi JST Sales America Inc.	VHR-2N	DigiKey	https://www.digikey.com/en/products/detail/jst-sales-america-inc/VHR-2N/608624	0.073	0.73	Cable-side mating housing.	
Protection	F1	1	5	Yes	5A cartridge fuse	Littelfuse	0217005.MXP	DigiKey	https://www.digikey.com/en/products/detail/littelfuse-inc/0217005-MXP/777549	0.436	2.18	1 on board plus spares.
Connector	1J14/J5	3	5	Yes	2-pin JST VH vertical PC JST Sales America Inc.	B2PS-VH	DigiKey	https://www.digikey.com/en/products/detail/jst-sales-america-inc/B2PS-VH/26555	0.16	0.8	Used for 12V JST input and two motor headers.	
Indicator	LED1	1	3	Yes	Green 1206 LED	Lite-On Inc.	LTST-C150GKT	DigiKey	https://www.digikey.com/en/products/detail/lite-on-inc/LTST-C150GKT/269216	0.15	0.45	Status LED.
Indicator	LED2	1	3	Yes	Red 1206 LED	Lite-On Inc.	LTST-C150EKT	DigiKey	https://www.digikey.com/en/products/detail/liteon/LTST-C150EKT/269215	0.13	0.39	Status LED.
Power	U3	1	2	Yes	12V to 3.3V buck conv Digiodes Incorporated	AP63203WU-7	DigiKey	https://www.digikey.com/en/products/detail/diodes-incorporated/AP63203WU-7/9858426	0.71	1.42	Main 3.3V regulator.	
Power	U1	1	1	Yes	12V to 6V power modul Texas Instruments	LMZ36107/NOPB	DigiKey	https://www.digikey.com/en/products/detail/texas-instruments/LMZ36107-NOPB/2673198	31.11	31.11	Motor and encoder supply.	
Motor Driv	U4/U5	2	3	Yes	Dual H-bridge motor dr Toshiba Semiconductor	TB67H450FNG-EL	DigiKey	https://www.digikey.com/en/products/detail/toshiba-semiconductor-and-storage/TB67H450FNG-EL/10130994	1.29	3.87	One per motor.	
Protection	Fuse holder	1	2	Yes	PCB fuse clip / holder	Schurter Inc.	3-143-050	DigiKey	https://www.digikey.com/en/products/detail/schurter-inc/3-143-050/25689195	2.31	4.62	Matches selected fuse format.
Passive	C1/C4	2	10	Yes	10 uF 25V X7R 1206 ca Murata Electronics	GRM31CR71E106MA1Z	DigiKey	https://www.digikey.com/en/products/detail/murata-electronics/GRM31CR71E106MA1Z/13904832	0.179	1.79	Input capacitors. Order extras.	
Passive	C2	1	3	Yes	100 uF 6.3V ceramic ca Murata Electronics	GRM31CRD80107ME4F	DigiKey	https://www.digikey.com/en/products/detail/murata-electronics/GRM31CRD80107ME4F/13904781	0.42	1.26	6V rail output capacitor.	
Passive	CB/C8/C9	3	12	Yes	22 uF 1206 ceramic ca Würth Elektronik	845012E-11	DigiKey	https://www.digikey.com/en/products/detail/wuerth-elektronik/845012E-11/22994394/5453421	0.44	6V and 3.3V rail output capacitors.		
Passive	CS/C7/C12/C13	4	12	Yes	0.1 uF 50V X7R 1206 ca Yageo	CRG320R/X7R98B104	DigiKey	https://www.digikey.com/en/products/detail/yageo-inc/CRG320R-X7R98B104/2833090	0.112	1.344	Bootstrap for local decoupling capacitors.	
Connector	J6/J7	2	4	Yes	4-position motor/encoc TE Connectivity AMP C	281695-4	DigiKey	https://www.digikey.com/en/products/detail/te-connectivity-amp-connectors/281695-4/1150102	0.11	4.04	Encoder headers.	
Switch	SW1	1	3	Yes	Momentary tactile push E-Switch	LT9320AF-000Q	DigiKey	https://www.digikey.com/en/products/detail/te-connectivity-amp-connectors/281695-4/1150102	0.9	2.7	Debug button.	
Passive	R3/R4	2	10	Yes	100 ohm 1206 1% resist Yageo	RC1206FR-07100RL	DigiKey	https://www.digikey.com/en/products/detail/yageo/RC1206FR-07100RL/728491	0.017	0.17	LED series resistors.	
Passive	R5/R6	2	10	Yes	10 kOhm 1206 1% resist Yageo	RC1206FR-0710KL	DigiKey	https://www.digikey.com/en/products/detail/yageo/RC1206FR-0710KL/728483	0.017	0.17	Pull-up resistors for DEBUG_BTN and EN_ESP.	
Passive	R1	1	10	Yes	6.8k Ohm 1206 1% res Yageo	RC1206FR-076K49L	DigiKey	https://www.digikey.com/en/products/detail/yageo/RC1206FR-076K49L/729022	0.011	0.11	LM23610 feedback upper resistor for 6V output.	
Passive	R2	1	10	Yes	1 kOhm 1206 resist Yageo	RC1206FR-071PK	DigiKey	https://www.digikey.com/en/products/detail/yageo/RC1206FR-071PK/4935356	0.02	0.2	LM23610 feedback lower resistor for 6V output.	
Magnetics	L1	1	3	Yes	6.8 uH power inductor KEMET	MPX1D1040L6R8	DigiKey	https://www.digikey.com/en/products/detail/diodes-incorporated/AP63203WU-7/9858426	0.71	2.13	Inductor for AP63203 buck regulator.	
Switch	Power source select	1	2	Yes	SPST source-select swi NKK Switches	100SP3T1B1M2QE	DigiKey	https://www.digikey.com/en/products/detail/nkk-switch/100SP3T1B1M2QE/3777888	2.95	5.9	Switch to control power between barrel jack and shared power rail	
Total										70.464		